

Mini-review

Regulatory T cells and infection: suppression revisited

Hans-Willi Mittrücker and Stefan H. E. Kaufmann

Department of Immunology, Max-Planck-Institute for Infection Biology, Berlin, Germany

Within the last few years, regulatory T cells (T_{reg} cells) have re-entered the stage of immunology. This astonishing revival is mainly due to recent studies on peripheral tolerance, which demonstrated the existence of T_{reg} cells and allowed the isolation and phenotypic and functional characterization of these cells. However, information on the role of T_{reg} cells in the control of immune responses against infections is still limited, although host defense represents the prime function of the immune system. In this overview, we summarize current knowledge on T_{reg} cells in immunity to infection, discuss potential functions of T_{reg} cells during infection, and finally, point to possible implications of T_{reg} cell functions for our understanding of fundamental aspects of host–pathogen interactions.

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1 Introduction

The existence of a suppressor T cell population was postulated several decades ago [1]. It was suggested that these suppressor T cells restrict induction or expression of effector T cells and thereby prevent or terminate exaggerated responses or, in the case of self-reactive T cells, autoimmune disease. According to this concept, suppressor T cells represent a crucial element in a regulatory network that controls immune responses and maintains peripheral tolerance. However, because of the failure to isolate and phenotypically characterize these lymphocytes, suppressor T cells became probably the most controversial topic in the field of immunology. Within recent years, the picture has changed dramatically. Suppressor T cells, now less provocatively renamed “regulatory” T cells (T_{reg} cells), have been isolated from mouse and man, and their suppressive or regulatory capacity has been demonstrated *in vitro* and *in vivo*. Overall, these observations define T_{reg} cells as central components of fundamental immune functions such as self tolerance, anti-tumor responses, and control of infection [2–5]. Here, we concentrate on the last of these components. We briefly describe phenotype and classification of T_{reg} cells; then we summarize observations on

T_{reg} cells in different infectious diseases and experimental infection models; finally we point to possible implications of T_{reg} cell functions for our understanding of fundamental aspects of host–pathogen interactions.

2 The phenotype and classification of T_{reg} cells

2.1 Distinct populations of T_{reg} cells have been defined

In spite of the experimental evidence for the existence of T_{reg} cells, the phenotype and mode of function of these cells still remain ill defined. On the basis of the experimental models used to analyze T_{reg} cells, they can be designated to two major cell populations: naturally occurring T_{reg} cells and induced T_{reg} cells, [6, 7].

2.2 Naturally occurring T_{reg} cells

Naturally occurring T_{reg} cells consist of $CD4^+$ T cells that mature in the thymus to regulatory T cells. They represent 5–10% of the peripheral $CD4^+$ T cell population and are characterized by the constitutive expression of CD25 and low expression levels of CD45RB [6, 7]. Additional surface molecules that have been associated with at least subpopulations of naturally occurring T_{reg} cells are GITR (TNFRSF18), CD152 (CTLA-4), CD103 (α_E -integrin)

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Abbreviations: TLR: Toll-like receptor T_{reg} cell: Regulatory T cell

and CD134 (OX40). However, none of these molecules, including CD25, is exclusive for T_{reg} cells, and there is evidence for $CD4^+$ T cells and even for $CD8^+$ T cells that express regulatory functions but lack CD25 and other T_{reg} marker molecules. In contrast, the expression of the transcription factor Foxp3 appears to be a unique and essential attribute of naturally occurring T_{reg} cells, and both mice and humans deficient in a functional Foxp3 protein suffer from severe autoimmune disease symptoms [8].

Naturally occurring T_{reg} cells need TCR signals for activation but, in contrast to regular effector T cells, these T_{reg} cells show only marginal proliferation and production of mitogenic cytokines following TCR stimulation *in vitro*. Naturally occurring T_{reg} cells appear to be mainly restricted by self antigens. However, examples for pathogen-specific T_{reg} cells also exist. Despite intensive research, the mechanisms underlying the suppressive functions of naturally occurring T_{reg} cells are still incompletely understood. *In vitro*, suppression depends on direct cell-to-cell contact. However, the surface molecules involved in this inhibition are still elusive. *In vivo*, suppressive cytokines, particularly IL-10 and TGF- β , appear as central mediators. In some experiments, the surface molecule CTLA-4 has been implicated in addition, since anti-CTLA-4 mAb reduce suppression [2, 3]. Targets for suppression are both APC and responding effector T cells, and there is evidence for suppression of components of the innate immune system. Depending on the targets of suppression and the environment and stage of immune response, naturally occurring T_{reg} cells could employ different suppressor mechanisms [2, 6, 7].

2.3 Induced T_{reg} cells

The second subtype of T_{reg} cells — induced T_{reg} cells — represents $CD4^+$ T cells that have acquired their suppressor activity during activation *in vitro* or *in vivo* [6, 7]. Induced T_{reg} cells are derived from $CD4^+CD25^-$ T cells and show variable expression of CD25 and other surface markers described for naturally occurring T_{reg} cells. Induced T_{reg} cells require TCR stimulation for induction of suppressive functions and demonstrate limited proliferation *in vitro*. Depending on the experimental model, suppression by induced T_{reg} cells is mediated via IL-10 and/or TGF- β . Induced T_{reg} cells show specificity for self or foreign antigens. According to the current concept, these cells are derived from regular $CD4^+$ T cells in the peripheral immune system. Critical determining factors for the development of induced T_{reg} cells include the type and differentiation status of the APC and the cytokine milieu during activation. Antigen presentation by immature DC in presence of IL-10 and/or TGF- β during T cell

activation apparently promotes differentiation of $CD4^+$ T cells into induced T_{reg} cells [4, 6].

In this review, we have divided T_{reg} cells into these two major subpopulations — naturally occurring T_{reg} cells and induced T_{reg} cells. Although this division can accommodate most experimental data, it is still hypothetical. Currently, the interrelation of the postulated T_{reg} cell populations remains unclear. Moreover, whether both populations consist of uniform lineages or have to be further subdivided into cell population with unique characteristics and functions needs to await further detailed analyses.

3 T_{reg} cells and infection

3.1 Regulation is a critical feature of immunity to infection

Currently, our knowledge on T_{reg} cells is mainly based on experimental models analyzing the regulation of immune responses in the context of self-tolerance and autoimmunity, graft and tumor rejection, and T cell homeostasis. One of the main functions of the immune system is the control of infection, and contact with a pathogen requires a strong and efficient response of the immune system to prevent harm for the organism. However, potent immune responses can be accompanied by severe side-effects with immunopathology as a final outcome. Thus, anti-pathogen responses need to be controlled adequately. It is therefore not surprising that the role of T_{reg} cells during the course of infection has attracted interest.

In retrospect, an abundance of results has described production of IL-10 and/or TGF- β , and presence of T cells secreting these cytokines both in experimental infection models and in individuals suffering from diverse infections. It is appreciated that during chronic infections, suppressive cytokines, in particular IL-10 and TGF- β , are produced and presumably involved in the regulation of the immune response and it has been postulated that these cytokines prevent exaggerated immune responses and pathology [9–13]. Of course, not every T cell that secretes IL-10 or TGF- β is a T_{reg} cell. However, it is likely that at least in some situations, re-evaluation of these studies will reveal a prominent role of T_{reg} cells in the control of chronic infections.

Beside these indirect pieces of evidence, a number of studies directly describe the presence and function of T_{reg} cells in various animal infection models as well as in human infectious diseases. These reports are based on analyses of a variety of pathogens, including viruses,

bacteria, fungi and parasites. In the following sections, we detail several studies of T_{reg} cell functions in the context of infectious diseases.

3.2 Models related to inflammatory bowel disease

Powrie and colleagues demonstrated that adoptive transfer of naive T cells into T cell deficient mice causes massive inflammatory disease in the recipients with the most prominent inflammatory manifestation in caecum and colon. Co-transfer of $CD4^+CD25^+CD45RB^{low}$ T cells suppressed development of intestinal inflammation and suppression was mediated by IL-10, TGF- β and CTLA-4. Intestinal inflammation was closely associated with the gut flora since mice that were kept under germ-free conditions failed to develop inflammatory disease [5]. Although these studies represent an inflammatory or autoimmune rather than an infection model (the response is triggered by the normal gut flora and $CD4^+$ effector T cells causing tissue damage are most likely directed against self antigens), these observations indicate that microbes can initiate inflammation and pathology, and that T_{reg} cells are able to prevent such detrimental responses.

On the basis of the studies of Powrie and colleagues, several groups have developed models in which inflammation is induced by oral infection of mice with defined bacterial species. In RAG-deficient mice, infection with *Helicobacter hepaticus* caused mild intestinal inflammation [14, 15]. Adoptive transfer of naive $CD4^+CD25^+CD45RB^{high}$ T cells or $CD4^+$ T cells from *H. hepaticus*-infected IL-10-deficient mice drastically enhanced inflammation and caused severe colitis in infected recipient mice. Co-transfer of $CD25^+$ or $CD25^-CD45RB^{low}CD4^+$ T_{reg} cells suppressed induction of intestinal inflammation by an IL-10-dependent mechanism [14, 15].

In a similar approach, T cell deficient *nu/nu* mice received $CD4^+$ T cells or $CD4^+$ T cells deprived of $CD25^+$ cells, and were orally infected with *H. pylori* [16]. Transfer of $CD25$ -depleted T cells significantly reduced gastric *H. pylori* colonization. However, this treatment also caused severe gastric inflammation, which was not observed in infected mice reconstituted with the unfractionated $CD4^+$ T cell population. Thus, T_{reg} cells can restrict T cell responses directed to or initiated by infection with defined bacterial species persisting in the gastrointestinal tract. T_{reg} cells thereby prevent severe inflammation and pathology caused by uncontrolled T cell responses. A recent study suggests a role for $CD4^+CD25^+T_{reg}$ cells in human *H. pylori* infection [17]. T_{reg}

cells isolated from chronically *H. pylori*-infected individuals suppressed *H. pylori*-specific T cell responses but failed to suppress responses to unrelated antigens. However, the significance of these T_{reg} cells for the pathology of human *H. pylori* infection remains to be determined.

3.3 *Bordetella pertussis*

T_{reg} cells have been isolated from lungs of mice infected with *B. pertussis* [18]. These T_{reg} cells express a $CD4^+CD25^+CTLA-4^{low}$ phenotype and secrete IL-10, IL-5, and in some cases TGF- β but not IFN- γ , IL-2 or IL-4 upon stimulation with *B. pertussis* antigens. *B. pertussis*-specific T_{reg} cells suppress T cell responses *in vitro* and restrict protective Th1 cell responses against *B. pertussis in vivo*. The virulence factor filamentous hemagglutinin (FHA) of *B. pertussis* preferentially induces T_{reg} cells. FHA inhibits production of IL-12 and stimulates production of IL-10 by DC. T cells activated by these DC preferentially differentiate into T_{reg} cells. This mechanism allows *B. pertussis* to suppress protective $CD4^+$ T cell responses in the infected host [18].

3.4 *Candida albicans*

Similar features have been observed after infection of mice with the fungus *C. albicans* [19]. Fungal infection induced a $CD4^+CD25^+T_{reg}$ cell population, which produced IL-4, IL-10 and TGF- β , showed poor proliferation *in vitro* and suppressed protective $CD4^+$ Th1 cell responses against *C. albicans in vivo*. In the absence of these T_{reg} cells, mice demonstrated enhanced protection against *C. albicans*, which was, however, accompanied by severe pathology. *C. albicans* directly promoted generation of T_{reg} cells in this model. *C. albicans* hyphae, but not yeast, induced the production of IL-10 by DC and immunization of mice with hyphae-loaded DC caused expansion of the $CD4^+CD25^+T_{reg}$ cell population.

3.5 *Pneumocystis carinii*

P. carinii is an opportunistic fungus for which a functional role of T_{reg} cells has been demonstrated. *P. carinii* causes pneumonia in immunocompromised human individuals and mice [20]. In RAG-deficient mice, *P. carinii* gives rise to a chronic infection initially with mild pathology. Adoptive transfer of $CD4^+$ T cells depleted of $CD25^+$ cells significantly reduced the pathogen load in infected recipients [21]. However, recipient mice also developed severe lung inflammation and fatal acute wasting disease. Co-transfer of $CD4^+CD25^+T_{reg}$ cells prevented the T cell mediated pathology, but also restricted the control of

infection. Thus, T_{reg} cells control the fate of infection with opportunistic microbes and determine whether infection transforms to disease or remains inapparent.

3.6 *Leishmania major*

Intradermal infection with *L. major* provokes strong local inflammation and accumulation of $CD4^+$ T cells in skin lesions and draining lymph nodes of mice. A considerable $CD4^+CD25^+$ T cell population has been demonstrated in skin lesions. These cells secrete IL-10 and suppress T cell responses *in vitro* — hallmarks of genuine $CD4^+CD25^+$ T_{reg} cells [22].

Further analyses revealed a fundamental role for these T_{reg} cells in control of *L. major*-specific T cell responses in both susceptible and resistant mouse strains [22–25]. Antibody-mediated depletion of $CD25^+$ T cells in susceptible BALB/c mice or transfer of $CD25$ -depleted $CD4^+$ T cells from BALB/c mice into SCID mice followed by intradermal *L. major* infection caused elevated parasite titers, exaggerated T cell responses with high levels of IL-4 and IFN- γ production, and further worsened the already severe inflammation and pathology observed in susceptible mice. Co-transfer of $CD4^+CD25^+$ T_{reg} cells into SCID mice suppressed the enhanced T cell responses and reduced the tissue inflammation [24, 25]. In contrast, depletion of $CD25^+$ T cells from resistant C57BL/6 mice improved parasite control following *L. major* infection [25]. Amelioration was also observed after transfer of $CD25$ -depleted $CD4^+$ T cells into RAG-deficient mice [22]. These mice developed smaller lesions and generated higher numbers of *L. major*-specific Th1 cells. In some animals, treatment even induced sterile eradication of parasites, which usually persist at low levels. Co-transfer of $CD4^+CD25^+$ T_{reg} cells prevented enhanced T cell responses and parasite eradication in an IL-10 dependent manner [22, 23].

Overall, results from the *L. major*-infection model indicate that T_{reg} cells efficiently suppress both $CD4^+$ Th2 and Th1 cell responses. Accordingly, T_{reg} cells deviate the immune response into a beneficial or detrimental direction depending on the overall condition of disease: in resistant mice they exacerbate, whereas in susceptible mice they ameliorate, the disease outcome.

3.7 *Listeria monocytogenes*

In contrast to most microbial infections, which are predominantly controlled by $CD4^+$ T cells, experimental listeriosis of mice serves as a model for $CD8^+$ T cell mediated anti-microbial protection [26]. Depletion of $CD4^+$ T

cells during secondary *L. monocytogenes* infection of mice enhanced listeria-specific $CD8^+$ T cell responses [27]. Further studies using anti-listeria vaccines demonstrated that $CD4^+CD25^+$ T_{reg} cells restrict the magnitude of secondary $CD8^+$ T cell responses. These results demonstrate that T_{reg} cells control $CD8^+$ memory T cell responses and suggest a central role for T_{reg} cells in the regulation of $CD8^+$ T cells during repeated or chronic infection.

3.8 Virus-infection models

There is also evidence for T_{reg} cells in the control of viruses. In the mouse model of herpes simplex virus infection, the protective $CD8^+$ T cell response was accompanied by heightened suppressive function of naturally occurring $CD4^+CD25^+$ T_{reg} cells [28]. These T_{reg} cells controlled the magnitude of the $CD8^+$ T cell response, since depletion of $CD4^+CD25^+$ T cells enhanced, whereas transfer of these cells reduced, the numbers and function of virus-specific $CD8^+$ T cells during the acute and the memory phase of the response.

Chronic infection of mice with Friend leukemia virus causes immune suppression manifested by impaired tumor rejection [29]. Infected mice show enhanced numbers of $CD4^+CD25^+$ T cells, which suppress T cell effector functions *in vitro*. Adoptive transfer of these T_{reg} cells impaired tumor rejection in recipient animals. Suppression, at least in the *in vitro* experiments, depended on TGF- β and CTLA-4.

3.9 Multicellular parasites

$CD4^+$ T cells demonstrating all hallmarks of T_{reg} cells have been isolated from human individuals infected with the nematode *Onchocercas volvulus* [30, 31]. These T cells produced IL-10 and TGF- β , and suppressed proliferation of Th1 and Th2 cells *in vitro*. Only a fraction of these T_{reg} cells was activated by *O. volvulus* antigens. The correlation of the frequency of IL-10- and TGF- β -secreting T cells and the outcome of infection implies a significant role of T_{reg} cells in the control of *O. volvulus*.

4 Functions of T_{reg} cells during infection

4.1 The delicate balance between anti-infective responses and immunopathology

The experiments summarized here emphasize involvement of T_{reg} cells in immunity against infection. The most obvious function of T_{reg} cells during infection is to regu-

late the delicate balance between the T cell responses that efficiently control the pathogen, and the exaggerated T cell responses that cause severe inflammation and tissue damage. This regulatory activity has been observed in virtually all types of infection in which T_{reg} cells have been studied. Alterations in number or function of T_{reg} cells can, on the one side, reduce inflammation but this is often accompanied by impaired pathogen control; on the other side, pathogen control can be improved, leading in some cases to sterile eradication of otherwise persisting pathogens, with the risk of more-severe or even fatal immunopathology. Thus, a critical function of T_{reg} cells is to dampen immune responses thereby preventing tissue damage even at the cost of impaired pathogen control.

Beside immunopathology, there are further reasons to limit the magnitude of immune responses, particularly during chronic infections. If unrestricted, the continuous activation of T cells by a persisting pathogen could lead to disproportional expansion of pathogen-specific effector T cell populations. A massive response focused on a single pathogen could impair adequate host responses to subsequent infections with unrelated pathogens. Thus, in the absence of T_{reg} cells, chronic infection could increase susceptibility to normally opportunistic microbes.

4.2 T_{reg} cells are predetermined to regulate immune responses during infection

To a certain degree, virtually all infections trigger inflammation and tissue damage. In this situation, APC present not only pathogen-derived antigens but also self antigens released from destroyed cells and thereby activate self-reactive T_{reg} cells as well as pathogen-reactive T_{reg} cells. Consequently, both self-reactive naturally occurring T_{reg} cells and self- or pathogen-reactive induced T_{reg} cells can become activated. T_{reg} cells possess further attributes that qualify them for the regulation of immune responses to infection. T_{reg} cells express receptors for pro-inflammatory chemokines and are attracted to sites of infection [32, 33]. In contrast to regular CD4⁺ T cells, T_{reg} cells are equipped with a wide spectrum of Toll-like receptors (TLR), including the LPS receptor TLR4 [34]. Stimulation of T_{reg} cells with TLR ligands such as LPS enhances suppressor activities. Thus, beside recognition of self or pathogen antigens on APC, T_{reg} cells can directly sense invading pathogens via TLR, and TCR- and TLR-mediated signals synergize in the induction of suppressive responses. However, stimulation of DC via TLR causes the release of soluble mediators that suppress the activity of T_{reg} cells [35]. The significance of these apparently contradictory observations remains to

be determined. Positive and negative regulation of T_{reg} cells via TLR could provide a means for adjusting T_{reg} cell functions to the site and/or the stage of the immune response.

4.3 Activation of T_{reg} cells: a potential immunoevasion strategy?

In numerous circumstances, induction or activation of T_{reg} cells during infection contributes to the protective host response. However, some studies also suggest that pathogens actively provoke the generation of T_{reg} cells thereby “misusing” these cells to evade the immune response. Several pathogens have been described that either express homologs of suppressive cytokines [36] or induce the expression of such cytokines in host cells. By a less-well understood process, pathogens can also interfere with DC differentiation either by blocking maturation or by inducing aberrant differentiation. Both mechanisms can cause generation of self- or pathogen-specific T_{reg} cells. Although initial findings support the concept, this topic clearly needs more-detailed analyses [18, 19].

4.4 Activation of T_{reg} cells: possible advantages for host and pathogen

Suppression of anti-pathogen responses by T_{reg} cells can be to the benefit of pathogen and host. During co-evolution over tens of thousands of years, pathogens causing chronic infections and their hosts have become highly adapted to each other. Therefore, chronic infection cannot be viewed solely as failure to generate an adequate immune response. Rather, it represents the outcome of a compromise between host and pathogen that respects the minimal requirements for both sides. We assume that T_{reg} cells are central regulators in the maintenance of this delicate balance.

The advantages for persisting pathogens are obvious: prolonged survival of a latently infected host, accompanied by disease outbreaks at later time-points, improves the spread of the pathogen. The advantages for the host of tolerating a chronic low-dose infection are less apparent. Analyses of the role of T_{reg} cells in the *L. major* infection models offer a possible explanation to this question [22, 23]. Reconstitution of RAG-deficient mice with CD25-depleted CD4⁺ T cells enhanced protection against *L. major* in mice such that levels of parasites were close to or below detection level. Co-transfer of CD4⁺CD25⁺ T_{reg} cells from wild-type mice impaired parasite control, causing chronic infection. T_{reg} cell function strictly depended on IL-10 because co-transfer of

CD4⁺CD25⁺ T cells from IL-10-deficient mice failed to affect protective immunity. Likewise, IL-10-deficient mice or wild-type mice treated with an antagonistic anti-IL-10R mAb showed enhanced protection, with sterile parasite eradication in some mice [22, 23]. However, in contrast to wild-type mice, IL-10-deficient mice and mice treated with anti-IL-10R mAb were not protected against secondary *L. major* infection and suffered from parasite titers similar to those in primary infected mice. Thus, infection that persists at a low level promotes the maintenance of immunologic memory. Whether this is due to continuous T cell triggering by pathogen-derived antigens or other mechanisms needs further investigation. Future studies will have to determine the significance of pathogen persistence for maintenance of immune memory in chronic human infections and the role of T_{reg} cells in this context.

4.5 Chronic infections, T_{reg} cells and the hygiene hypothesis

In the last decades, industrialized nations have witnessed a dramatic increase in the prevalence of autoimmune disorders, including diseases connected to deregulated Th1 cell responses, such as inflammatory bowel disease and autoimmune diabetes, as well as Th2-cell-associated allergic diseases. In parallel, these populations have experienced a decline in infectious diseases, mainly because of improved hygiene standards and the development of efficient intervention regimes. The “hygiene hypothesis” connects these two developments by proposing that increased incidences of autoimmune diseases are due to diminished pathogen pressure on the immune system. Although this hypothesis is far from being fully proven by experimental and epidemiologic data, it offers a plausible explanation for the observed increases in autoimmune diseases (reviewed in [37, 38]).

According to current knowledge, T_{reg} cells could play a critical role in this concept. On the one hand, T_{reg} cells are involved in the control of peripheral tolerance. In several experimental models, alterations in numbers or activities of T_{reg} cells cause severe autoimmune disorders. On the other hand, infections — particularly chronic infections — lead to changes in numbers and activities of T_{reg} cells. Furthermore, at least a subpopulation of T_{reg} cells is self-specific. Thus, T_{reg} cells generated or activated during infection are not restricted to immune responses against the homologous pathogen, but can also influence unrelated aspects of immune homeostasis. Consequently, diminishing the pressure from infectious agents and thereby the activation state of T_{reg} cells will also relieve control on self-reactive T cells and increase the risk of autoimmune disorders.

Although this concept is attractive, it is in most parts based on indirect observations and thus largely speculative. Key issues that remain to be investigated include whether the status of T_{reg} cells differs between individuals suffering from chronic or repeated infections compared with less-frequently infected individuals, and whether elevated numbers or activities of T_{reg} cells prevent or restrict autoimmune diseases. In case this concept holds true, modulation of T_{reg} cell activity could offer a means for therapy of autoimmune and allergic disorders.

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Correspondence: Hans-Willi Mittrücker, Department of Immunology, Max-Planck-Institute for Infection Biology, Schumannstrasse 21/22, 10117 Berlin, Germany
 Fax: +49-30-28460-501
 e-mail: mittruecker@mpiib-berlin.mpg.de